ERC Starting Grant 2026

Part B1

Flexible Intelligence through Neurocomputational Design

FIND

Principal Investigator:	Dr. Mikkel Elle Lepperød
Host Institution:	Simula Research Laboratory, Norway
Duration:	60 months
Primary Panel:	PE6 — Computer Science and Informatics

Abstract

Current AI systems cannot selectively store and retrieve information across thousands of timesteps—a capability biological brains achieve routinely through distributed memory systems. Transformers forget beyond their context window; state space models compress everything into fixed-size state; and memory-augmented networks lack causal models to decide what is worth storing. FIND develops **PAM** (**P**redictive **A**ctive temporal coding with distributed **M**emory), a novel brain-inspired architecture where modules of neurons read and write to local long-term memory, combining temporal predictive coding, hyperdimensional computing, and active inference into a unified framework.

PAM combines three mechanisms: (1) *Distributed modular specialization* where each module maintains dedicated content-addressable memory; (2) *Temporal predictive coding* where prediction errors drive memory updates and learning; and (3) *Active inference* where action selection minimizes uncertainty. Our preliminary results demonstrate PAM's viability beyond toy benchmarks: 100% recognition accuracy on CORe50 (50 objects) with **zero additional forgetting by construction**—nonparametric storage cannot overwrite parameters; **98.2% online continual learning accuracy** across 120K samples in 55 seconds—12× faster than gradient-based methods; and **24.7% exact match on ARC** (the Abstraction and Reasoning Corpus) with only 1.47M parameters and no pretraining, approximately 2× GPT-4o's 12% at >1000× fewer parameters.

FIND will develop PAM through three work packages: **WP1** establishes temporal predictive coding with distributed memory for continual learning and long-range temporal reasoning; **WP2** extends this to compositional reasoning and program synthesis through neurosymbolic HDC integration; and **WP3** stress-tests the system in demanding open-ended environments (Memory Maze, Minecraft, Factorio) requiring planning and adaptation across >10,000 timesteps.

FIND: Flexible Intelligence through Neurocomputational Design

1 Part I of the Scientific Proposal

We live in time. Memory bridges past and present, enabling agents to learn from experience and make informed decisions. Humans develop internal world models for planning and reasoning; replicating this in machines is the next step toward flexible AI [1]. Current AI struggles to generalise [2], lacking long-term memory [3], causal understanding [4], and life-long learning without catastrophic forgetting [5].

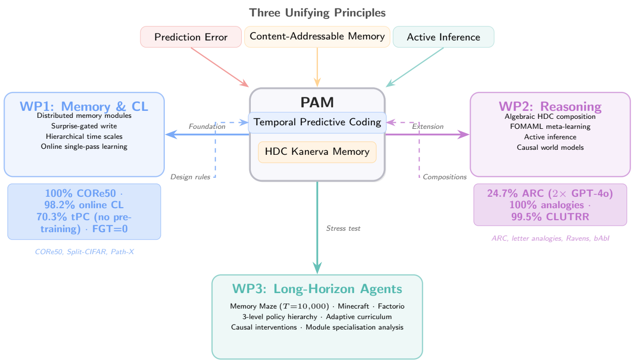

Figure 1: Conceptual overview of FIND. PAM (centre) unifies temporal predictive coding and HDC-based content-addressable memory. Three work packages build outward: **WP1** establishes the memory and continual learning foundation; **WP2** extends with compositional reasoning and causal world models; **WP3** stress-tests in demanding long-horizon environments. Preliminary results (coloured boxes) validate each WP's core claims.

Consider an agent managing a Factorio production chain (Figure 2). It discovers copper wire requires copper plates (step 2,000) and circuits require wire plus iron (step 8,000), then encounters a blueprint requiring 50 circuits at step 15,000. It must recall the dependency chain across >10,000 timesteps, reason causally about bottlenecks, and adapt its layout. Current systems cannot do this: transformers cannot attend across 10,000 steps [6]; SSMs compress dependencies into fixed-size state [7]; MANNs lack causal models to identify bottlenecks [8]. The challenge is to achieve this *learned from raw experience*, without a pretrained model or hand-engineered memory plumbing. **How can systems intelligently select what to retain, discard, and retrieve to learn across long temporal sequences?**

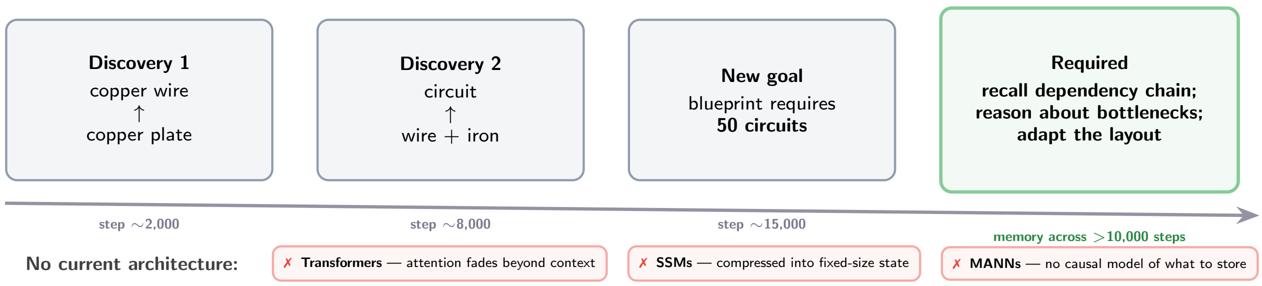

Figure 2: The capability gap FIND addresses. An agent discovers dependencies thousands of steps apart and must later recall and compose them under a new goal. This requires selective storage, content-addressable recall, and causal reasoning across >10,000 timesteps—beyond current architectures (bottom).

1.1 Ground-breaking nature and impact

FIND proposes a paradigm shift drawing inspiration from biological memory systems. We develop **PAM** (**P**redictive **A**ctive temporal coding with distributed **M**emory), a flat modular architecture where complexity arises from learned routing and memory interactions, operating through:

- Specialized modules with dedicated Kanerva memory [9, 10]
- HDC symbolic representations [11] retained in memory
- Learned orchestration for dynamic information routing
- Temporal predictions via predictive coding; action selection via active inference [12, 13]

FIND's novelty lies at the intersection of three established fields in a configuration no prior system has achieved simultaneously:

- **Temporal predictive coding** [14] with **distributed external memory**—not just recurrent state: prior tPC models store context solely in hidden activations; FIND externalises memory into content-addressable HDC slots persistent across thousands of timesteps.
- **Memory-augmented networks** [8] with **biologically-grounded iterative inference**—not single-pass read/write: prior MANNs use a single forward pass to address memory; FIND uses

iterative predictive-coding inference to resolve ambiguous queries, yielding a 1.75× recall advantage at σ=0.4.

- **Hyperdimensional computing** [11] **embedded as the storage format of a gradient-trained system**—not as a standalone symbolic method: prior HDC systems operate outside neural training loops, and even NVSA [15], which couples vector-symbolic reasoning to a trained perception network, is a single-task inference pipeline without persistent temporal memory, continual learning, or action; FIND makes HDC the storage substrate of a continual, temporal system, integrating binding directly into the memory update.

Table 1 situates FIND against the state of the art. No existing system combines all four capabilities.

Table 1: Positioning of PAM against existing approaches. Only PAM combines external distributed memory, iterative inference, symbolic binding, and active inference in a single end-to-end trainable architecture.

System	External Memory	Iterative Inference	Symbolic Binding	Active Inference
NTM / DNC [8, 16]	✓	—	—	—
Kanerva Machine [9]	✓	—	—	—
Temporal PC [14]	—	✓	—	—
HDC systems [11]	—	—	✓	—
S4 / Mamba [7]	— (compressed state)	—	—	—
Titans [17]	✓	—	—	—
NVSA [15]	—	—	✓	—
Active PC [13]	—	✓	—	✓
PAM (FIND)	✓	✓	✓	✓

1.2 State of the art, knowledge needs, and project objectives

Temporal memory and predictive coding

A key challenge in AI is creating memory systems comparable to the brain. RAG [18, 19] and agentic systems like MemGPT [20] manage context through external plumbing, but these sidestep bottom-up lifelong learning and struggle with out-of-distribution scenarios [2].

The core challenge is separating memory storage from computation [21, 22]. MANNs—NTMs [16], DNCs [8], Kanerva Machines [9]—address this partially. HDC extends sparse distributed memories with symbolic binding [11, 23]. Long-context transformers (Infini-Attention [24]) extend windows but do not solve selective storage. Modern SSMs—Mamba-2 [25], xLSTM [26]—achieve efficient sequence processing but compress into fixed-size state.

Predictive coding (PC) minimizes prediction errors between expectations and inputs [12, 27]. Extensions to active [13] and temporal PC [14] enable biologically plausible cognitive maps. Test-time training (TTT) [28] shares PAM's iterative adaptation principle but modifies the full model rather than memory slots. Titans [17] is closest in spirit: it gates test-time writes to a long-term memory by surprise—sharing PAM's write-gating principle—but stores into a monolithic parametric memory without symbolic binding, compositional retrieval, or action selection. **However, current PC models rely on Markov assumptions, unable to handle long-term credit assignment.**

Causal reasoning, world models, and compositional understanding

A causal world model represents generative mechanisms enabling counterfactual reasoning [4, 29]. Recent models—JEPA [30], DINO-WM [31], GAIA-1 [32]—show promise, but current benchmarks operate at irrelevant timescales: 2000 steps [33] vs. desert ants maintaining memories across 200 000 steps [34].

A complementary gap is *compositional reasoning*: factorizing knowledge into reusable subspaces for novel recombinations. The ARC benchmark [35] exemplifies this—tasks require recognising that transformations *compose* algebraically. FIND addresses this through HDC binding (bind, unbind, superposition) within the same space used for memory.

Cognitive map learning has been pursued in neuroscience [36] and AI [37], including by our lab [38–41], but these lack a unified theory of perception, cognition, and action—the gap FIND addresses.

Adaptive learning

Life-long learning requires managing information without catastrophic forgetting. Current approaches span regularization, replay, and Bayesian methods [5, 42]; we have explored generative stability [43] and plasticity [44]. FIND addresses this through HDC [11]: binding memories as compositional vectors enables fast retrieval and symbolic queries without retraining.

1.3 Research strategy and objectives

Central hypothesis: Predictive coding over content-addressable distributed memory slots is sufficient for selective long-horizon recall—the ability to store, retrieve, and compose information across thousands of timesteps without catastrophic forgetting—and achieves it where compressed-state methods cannot without unbounded state growth. We decompose this into three testable sub-hypotheses:

- **H1 (WP1):** Non-parametric HDC-structured memory is immune to parameter-overwriting forgetting *by construction*—a guarantee gradient-based methods do not offer—while maintaining competitive recognition accuracy; our CORe50 results (100% recognition, 98.2% online CL accuracy) confirm the accuracy side.
- **H2 (WP2):** Compositional binding in HDC space enables zero-shot generalisation to novel transformation compositions, with distributed memory as a load-bearing component of test-time program synthesis—PAMNCA ablations attribute 30–43% of performance to memory, growing with compute budget.
- **H3 (WP3):** Memory-augmented agents outperform compressed-state agents (SSMs, Transformers) specifically on tasks requiring recall across >5,000 timesteps.

We address this through three objectives and corresponding work packages:

- **Objective 1 / WP1: Temporal Predictive Coding with Distributed Memory.** Develop the foundational PAM architecture that processes information over different time scales through temporal predictive coding with distributed memory modules.
- **Objective 2 / WP2: Causal Reasoning and Compositional Understanding.** Extend PAM with active inference, neurosymbolic integration, and subspace-based compositional reasoning for causal world model learning.
- **Objective 3 / WP3: Long-Horizon Adaptation in Open-Ended Environments.** Stress-test the complete PAM system in demanding environments (mazes, Minecraft, Factorio) that require sophisticated planning, resource management, and long-horizon adaptation.

Theoretical approach. PAM combines three mechanisms:

1. *Distributed modular specialization:* Specialized modules with dedicated Kanerva memory [9], routed via Gumbel-softmax program induction iteratively over n stages.
2. *Temporal predictive coding:* Prediction errors drive memory updates and model learning.
3. *Active inference:* Action selection minimizes uncertainty, enabling efficient exploration and causally relevant memory storage.

PAM uses content-addressable memory with retrieval via similarity between z_t^m and stored basis vectors h_i^m, with exponential maps [45] for closed-form Riemannian updates. HDC [11, 45] bridges neural and symbolic approaches, enabling compositional reasoning through binding operations (Figure 3).

Preliminary results. The BioAI group has developed FIND's foundations through neurocomputational memory research: grid/place/border cells emerge spontaneously in AI agents [38], orthogonal remapping encodes context [39–41], NCA enables self-organizing module communication [46], one-shot memory transfer [47], and abstract reasoning [48]. The tPC-KM system achieves 98.1% exact recall at σ=0.4 (1.75× over feedforward) and 3.75× less forgetting than tPC-alone (Figure 4).

WP1: Memory and continual learning. Beyond the MNIST benchmark, PAM achieves **100% recognition accuracy** on CORe50 (50 real-world objects, T=10,000 steps) with both pretrained and raw pixel features. On the official CORe50-NI online continual learning protocol (test-then-train, single pass, no task boundaries), PAM achieves **98.2% online accuracy** across 8 experiences (120K samples) in 55 seconds—12× faster than gradient-based replay methods—with positive backward transfer and *zero additional forgetting* as experiences accumulate: non-parametric storage cannot overwrite parameters, so accuracy is bounded only by superposition capacity, which scales predictably with memory dimension [10]. With PAM's own tPC encoder (zero pretraining), the system reaches **70.3% per-experience accuracy** while learning both representations and memory online—the only continual learning system that operates without a pretrained backbone. The gap to the frozen-feature setting reflects *representation drift*: online-learned keys diverge from those stored earlier,

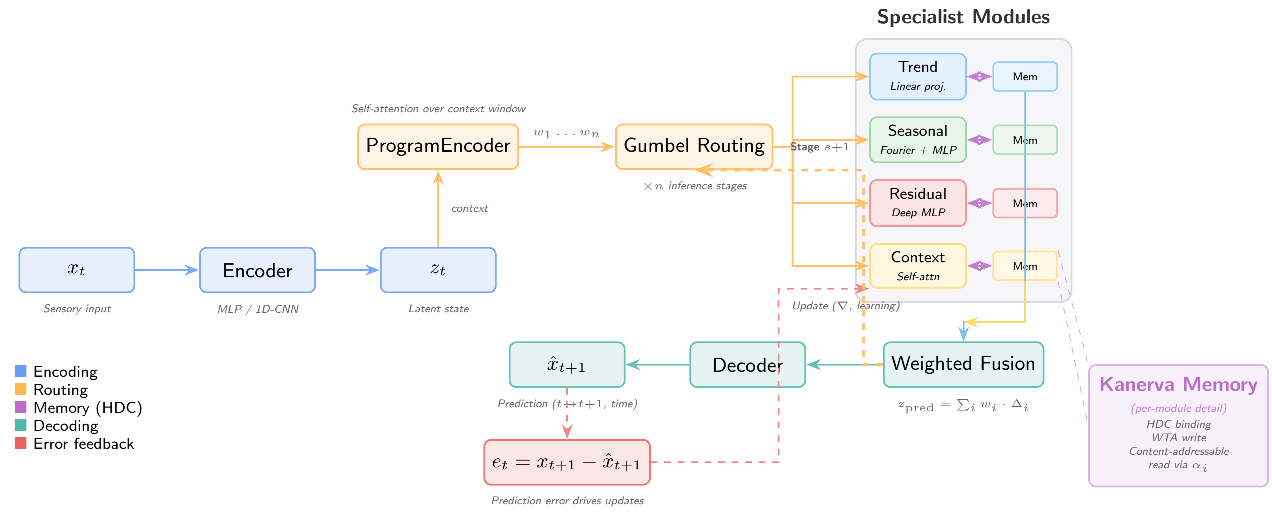

Figure 3: PAM architecture. Sensory input x_t is encoded into latent state z_t. A ProgramEncoder infers routing weights via self-attention over the context window. Gumbel-softmax routing selects among four specialist modules (Trend, Seasonal, Residual, Context), each with dedicated Kanerva memory using HDC binding for content-addressable read/write. The routing is applied iteratively over n stages. Module outputs are fused into a prediction $\hat{x}_{t+1}$; prediction errors drive parameter updates. The module inventory is task-dependent (shown here for temporal forecasting).

degrading content-addressable retrieval—the central open problem for episodic memories keyed by learned embeddings, which WP1 addresses with a key-reconsolidation mechanism.

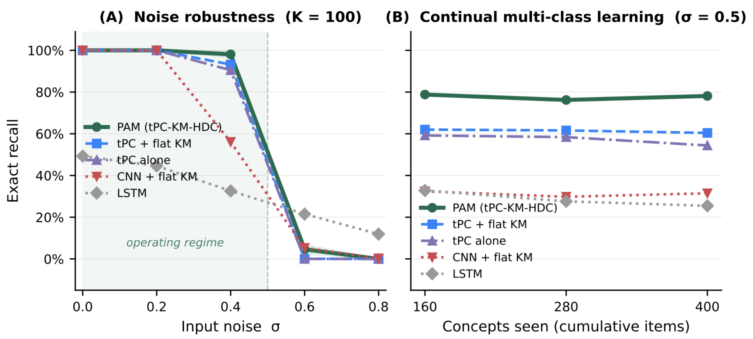

Figure 4: Preliminary PAM results (WP1 foundation). (A) Exact recall vs. input noise σ (K=100 items, 20 trials). Iterative inference (PAM, tPC-KM) outperforms feedforward encoding (CNN-KM) by 1.75× at σ=0.4. Beyond σ≈0.5 (shaded band), exact recall collapses for all content-addressable methods—a capacity/SNR phase transition predicted by HDC crosstalk analysis, since crosstalk in superposed hypervectors grows with stored items; deriving this boundary analytically is a T1.2 target (Part II). (B) Exact recall across three sequential class phases (σ=0.5). PAM maintains ≈78% recall across all phases; HDC binding prevents inter-class interference as new concepts accumulate, reducing forgetting 3.75× versus tPC alone.

WP2: Compositional reasoning. We validated PAM's program synthesis capabilities on the Abstraction and Reasoning Corpus (ARC) [35]. Our PAMNCA system—a neural cellular automaton with distributed PAM memory (1.47M parameters)—achieves **24.7% exact match** (321/1302 episodes) through test-time training, with no pretrained backbone. The significance is the efficiency frontier, not the absolute score: large program-synthesis ensembles now exceed this number using 10^5–10^6× more compute, whereas PAMNCA doubles GPT-4o's 12% at >1000× fewer parameters, learning each task from its support examples alone. Memory ablations show the distributed memory module contributes 30–43% of performance, with its contribution *growing* with compute budget. FOMAML meta-learning provides a 2.5× effective compute speedup, validating T2.2's meta-learning approach. Separately, our HDC system achieves 100% on direct analogies (with 200 distractors), 99.5% on CLUTRR 2-hop relational reasoning, and 100% on Raven's Progressive Matrices (NVSA [15] reaches similar accuracy on Raven's; PAM's distinction is that the same memory substrate also serves continual temporal recall rather than a dedicated reasoning pipeline).

PAM makes two testable predictions unique to its architecture: (1) non-parametric memory guarantees zero *additional* forgetting by construction (recall bounded only by superposition capacity) while maintaining recognition accuracy—our CORe50 experiments confirm 100% recognition versus the S4D baseline's 44.5% re-encounter accuracy; (2) distributed memory is load-bearing for program synthesis

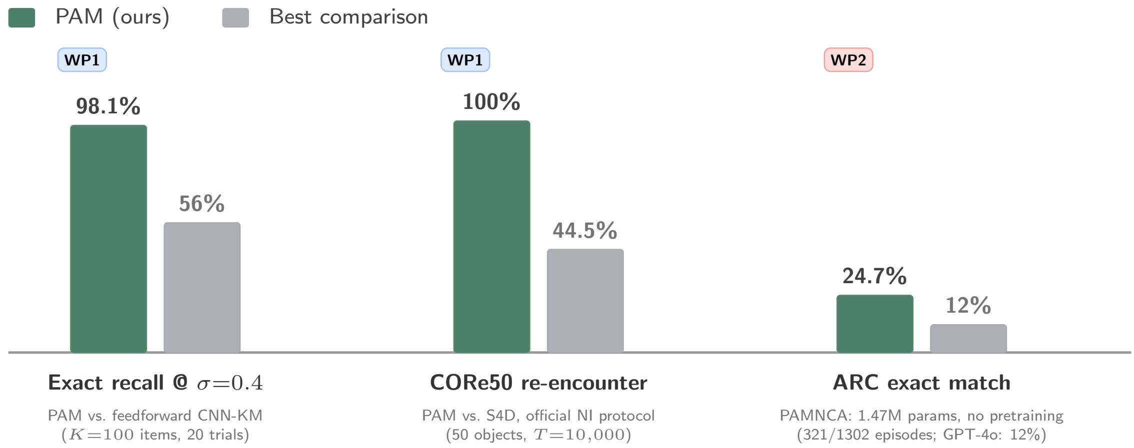

Figure 5: Headline preliminary results. PAM against the strongest comparison on each benchmark: noisy sequence recall at σ=0.4 (WP1 foundation), CORe50 re-encounter recognition after T=10,000 steps (WP1), and ARC program synthesis via test-time training (WP2).

via test-time adaptation, with its contribution growing with task complexity—our PAMNCA ablations support this (30% at TTT=500, 43% at TTT=1000).

1.4 Major expected outcomes

Science: A theoretical framework unifying predictive coding, distributed memory, and active inference beyond Markov assumptions. Understanding of how modular specialization emerges from predictive principles.
Technology: PAM as open-source framework with benchmarks spanning forecasting (ETT, Weather), long-range memory (Path-X, Memory Maze), and continual learning. Orchestration algorithms with reproducible evaluation suites.
Society: Foundations for AI with principled memory applicable to healthcare (patient timelines), industrial automation (Factorio-style planning), and environmental monitoring.
Success criteria: (i) $\geq$90% recall at σ=0.3 and MSE within 10% of PatchTST (WP1); (ii) >40% ARC exact match at <10M parameters without a pretrained backbone, extending current 24.7% (WP2); (iii) >10 pp over DQN on Memory Maze at T>5,000 (WP3). Team: PI (50%), one postdoc (ML systems), two PhDs; 4-tier benchmark hierarchy (Part II).

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2 Curriculum Vitae and Track Record

Personal details

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Education and key qualifications

2020	**Ph.D.** Neuroscience, Faculty of Medicine, University of Oslo, Norway
2014	**M.Sc.** Applied Mathematics, Norwegian University of Life Sciences, Norway

Current and relevant previous positions

2026–	**Senior Research Scientist** (equiv. associate professor), Simula Research Laboratory
2023–	**Adjunct Associate Professor** (20%), Dept. of Physics, University of Oslo
2022–2026	**Research Scientist** (equiv. assistant professor), Simula Research Laboratory
2020–2022	**Postdoctoral Fellow**, Dept. of Biosciences, University of Oslo

PhD supervisor: Prof. Torkel Hafting and Prof. Marianne Fyhn (University of Oslo).
Postdoctoral mentor: Self-directed (group leader from 2020).
Career breaks: Parental leave (5 months in 2017, 5 months in 2020).

Research outputs

The following ten outputs demonstrate my trajectory from experimental neuroscience to neuro-inspired AI, establishing the foundations for FIND:

1. **Lepperød ME** et al. *Optogenetic pacing of medial septum parvalbumin-positive cells disrupts temporal but not spatial firing in grid cells.* Science Advances, 2021. *Role: First author, conceived and led the study. Significance: Established causal evidence for temporal coding in the entorhinal cortex, directly motivating the temporal predictive coding component of PAM.*
2. **Lepperød ME** et al. *Inferring causal connectivity from pairwise recordings and optogenetics.* PLoS Computational Biology, 2022. (With K. Kording.) *Role: First author. Significance: Developed quasi-experimental methods for causal inference in neural circuits, grounding FIND's causal reasoning objectives.*
3. Schøyen VS*, Pettersen MB*, . . . , **Lepperød ME**. *Coherently remapping toroidal cells but not grid cells are responsible for path integration in virtual agents.* iScience, 2023. *Role: Senior/-corresponding author, supervised the project. Significance: Showed that AI agents spontaneously develop biologically consistent spatial cells, establishing our lab's cognitive map expertise.*
4. Pettersen MB, Schøyen VS, Malthe-Sørenssen A, **Lepperød ME**. *Decoding the cognitive map: Learning place cells and remapping.* eLife, 2024. *Role: Senior/corresponding author. Significance: Demonstrated context-dependent memory through orthogonal remapping, key to FIND's modular memory design.*
5. Pettersen MB, Rogge F, **Lepperød ME**. *Learning place cell representations and context-dependent remapping.* NeurIPS, 2024. *Role: Senior author. Significance: Extended cognitive map learning to context-switching, directly informing WP2's compositional reasoning.*
6. Kvalsund MK, Ellefsen KO, Glette K, Pontes-Filho S, **Lepperød ME**. *Sensor movement drives emergent attention and scalability in active neural cellular automata.* Neural Networks, 2026. *Role: Senior/corresponding author. Significance: Demonstrated NCA-based inter-module communication with active vision—the primary orchestration mechanism for PAM.*
7. Guichard E, Reimers F, Kvalsund MK, **Lepperød M**, Nichele S. *EngramNCA: A neural cellular automaton model of memory transfer.* ALife, 2025. *Role: Co-supervisor. Significance: Showed NCA-based one-shot memory transfer between modules, directly enabling PAM's distributed memory architecture.*

8. Guichard E, Reimers F, Kvalsund MK, **Lepperød M**, Nichele S. *ARC-NCA: Towards developmental solutions to the Abstraction and Reasoning Corpus.* ALife, 2025. Featured on Machine Learning Street Talk podcast. *Role: Co-supervisor. Significance: Demonstrated abstract reasoning through developmental self-organization, informing WP2's compositional reasoning.*
9. Schøyen VS et al. *Hexagons all the way down: Grid cells as a conformal isometric map of space.* PLOS Computational Biology, 2025. *Role: Senior/corresponding author. Significance: Provided geometric theory of neural representations, informing HDC integration in PAM.*
10. Danielli K, **Lepperød ME**. *A bio-inspired minimal model for non-stationary K-armed bandits.* Biological Cybernetics, 2026. *Role: Senior/corresponding author. Significance: Demonstrated bio-inspired adaptive learning under non-stationarity, directly relevant to WP3's open-ended adaptation.*

Significant peer recognition

- **NORA Award for Early Career Investigator** (2025)—National recognition for contributions to NeuroAI
- **Simula ERC Incubator** (2022)—Selected for institutional support program for ERC candidates
- **Wharton Executive Education** (2022)—"High-Potential Leaders: Accelerating Your Impact"
- **Invited speaker**: University of California San Diego (2022), AI Week Oslo Met (2023), Keynote at Young Researchers in NeuroAI (2023), Simula Bio-inspired Computing Seminar (2024)
- **Workshop organizer**: 4-day NeuroAI workshop on Hurtigruten with 20 international speakers (2024); co-organizer of 3+ NeuroAI workshops connecting national and international labs (2023–2024)
- **Reviewer**: Nature Machine Intelligence, PLoS Computational Biology, eNeuro
- **Board member**: NORA Special Interest Group for NeuroAI Norway
- **Public outreach**: Brain Inspired podcast (2024), Machine Learning Street Talk podcast (2025), NRK interview (2021), opinion pieces in digi.no and forskning.no

Project leadership and supervision

Group leader, BioAI group at Simula (2020–present): 4.5 MNOK/year, managing 3 PhD students, 1 postdoc. Built the group from inception to a productive research unit with publications at NeurIPS, eLife, Science Advances, and Nature Communications.
Acting co-PI, Bio-inspired neural networks for AI application (2020–2026): 16 MNOK RCN-funded project, University of Oslo. Co-supervised 3 PhDs.
Supervised to completion: 2 PhD students (V. Schøyen, K. Holzhausen), 8+ MSc students (including graduates now at Stanford PhD program and UiO PhD program).

Research software and ML systems engineering

PAM / PAMNCA codebases: Large-scale ML engineering behind the preliminary results in this proposal: systematic GPU-cluster experimentation (∼15K GPU-hours on A40s), automated result parsing from 150+ log files, reproducible evaluation pipelines, and benchmarking across 5 real-world domains (CORe50, MIT-BIH, ESC-50, VoxCeleb, embodied navigation) using official online continual-learning protocols. [TODO: repository URL and open-source release status]
Community research infrastructure: Co-developed NEST simulator modules within the Human Brain Project; co-developed the Exdir hierarchical data format and the Expipe experiment-management platform for reproducible neuroscience pipelines. [TODO: adoption indicators: downloads/stars/citing labs for Exdir and Expipe]
Benchmark contributions: Evaluation on official public protocols (CORe50-NI, ARC). [TODO: public leaderboard entries or benchmark submissions, if any]
Team profile to complement the PI: The FIND postdoc will be recruited from the ML systems community (target profile: PhD from an ICLR/ICML-publishing lab with large-scale training experience) to complement the PI's neuroscience background and own the WP2–3 scaling infrastructure. [TODO: 1–2 sentences on candidate pipeline, e.g. named ELLIS/NORA contacts]

Additional information

Community leadership: Lead role establishing NeuroAI as focus field in the Norway–US Ministry of Education collaboration (2022). Developed and taught the NeuroAI course at University of Oslo (2023).